Network Mini-project

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We analyze various properties of the all-or-none model of network growth.

I. INTRODUCTION

Network growth models are central to our understanding of complex systems in the real world. The Barabási–Albert (BA) model introduced preferential attachment as a mechanism for generating power-law degree distributions, while copying models propose an alternative microscopic rule in which new nodes replicate the connections of existing ones.

In this work we analyze the *all-or-none* model. At each growth step a new node is added and connected to a target chosen uniformly at random, and then it connects to *all* neighbors of the target with probability p, or to *none* of them with probability $1-p$. We show that despite its simplicity, the model produces power-law degree distributions with exponent $1/p$, an N-independent clustering coefficient $C = 3p/(1+2p)$, and logarithmic average distances, making it a small-world network for every $p \in (0,1)$. Additionally, we will also investigate the model's assortative nature, as well as its consensus dynamics.

II. MODEL DEFINITION

The network begins with a single node ($N = 1$). At each growth step ($N \to N+1$):

1. A new node v is introduced.
2. A target node u is chosen uniformly at random from the existing N nodes.
3. An edge (v, u) is added.
4. With probability p, edges (v, w) are added for every neighbor w of u; with probability $1-p$, no further edges are added.

III. NUMBER OF EDGES

Let M denote the expected number of edges in the network, and write M_N to specify when the network has N nodes. At step $N \to N+1$, a target u is chosen uniformly from the N existing nodes. The expected number of new edges is $1 + p\,k_u$, where k_u is the degree of u. Note that by the handshake lemma, the average value of k_u over all choices of u is $2M_N/N$. It follows that,

$$M_{N+1} = \left(1 + \frac{2p}{N}\right) M_N + 1. \tag{1}$$

This is a first-order linear recurrence. A particular solution of the form $\alpha_M N$ yields $\alpha_M = 1/(1-2p)$ for $p \neq 1/2$. The homogeneous solution is $\mathcal{A}_N = \Gamma(N+2p)/[\Gamma(1+2p)\,\Gamma(N)]$, satisfying $\mathcal{A}_{N+1}/\mathcal{A}_N = (N+2p)/N$. Imposing $M_1 = 0$ now yields

$$M = \frac{1}{1-2p}\left(N - \frac{\Gamma(N+2p)}{\Gamma(1+2p)\,\Gamma(N)}\right), \tag{2}$$

for $p \neq 1/2$. Since $\Gamma(N+2p)/\Gamma(N) \sim N^{2p}$ as $N \to \infty$, two regimes emerge: in the *sparse* regime ($p < 1/2$), the growth $M \sim \frac{N}{1-2p}$ is linear, whereas in the *dense* regime ($p > 1/2$), the growth $M \sim \frac{N^{2p}}{(2p-1)\,\Gamma(1+2p)}$ is superlinear.

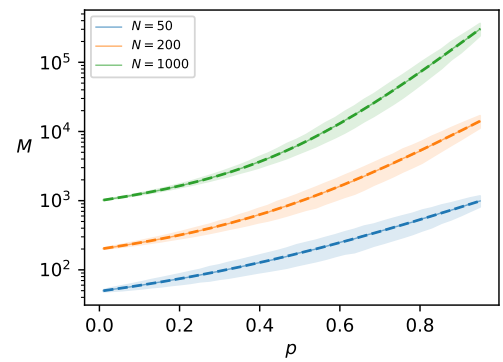

FIG. 1: Expected number of edges M (log scale) as a function of p for $N = 50, 200, 1000$ with 10^3 trials. Shaded regions show one standard deviation. The dashed lines are the analytical prediction (2), fitting the simulation results perfectly across all tested network sizes.

IV. DEGREE DISTRIBUTION

Denote by $f(k, i, N)$ the probability that node i has degree k when the network has N nodes. At step $N \to N+1$, degree k increases by one with probability $(1+pk)/N$, because either the node is chosen as target (probability $1/N$) or a neighbor of it is chosen and the "all" event occurs (probability pk/N). Thus,

$$f(k, i, N+1) = \frac{1+p(k-1)}{N} f(k-1, i, N) + \left(1 - \frac{1+pk}{N}\right) f(k, i, N). \tag{3}$$

Assume that $p < 1/2$ so that the expected number of edges M is linear in N, by the results of Section III. This allows us to seek the asymptotic degree distribution

$P_k = \lim_{N\to\infty} n_k(N)/N$, where $n_k(N) = \sum_{i=1}^{N} f(k,i,N)$ is the expected number of nodes with degree k in the N-node network. Note that the new node enters with degree 1 with probability $1-p$ and with degree k with probability $p\, n_{k-1}(N)/N$ for $k \geq 2$. Hence, summing (3) over the N existing nodes and adding the new node yields

$$n_k(N+1) = \frac{1+p(k-1)}{N}\, n_{k-1}(N) + \left(1 - \frac{1+pk}{N}\right) n_k(N) + (1-p)\,\delta_{k,1} + \frac{p\, n_{k-1}(N)}{N}. \tag{4}$$

Substituting $n_k(N) = NP_k$, we have $(2+pk)\,P_k = (1+pk)\,P_{k-1} + (1-p)\,\delta_{k,1}$. This yields $P_1 = (1-p)/(2+p)$ and, for $k \geq 2$, we have $P_k = \frac{1+pk}{2+pk}\,P_{k-1}$, which has the closed form

$$P_k = \frac{1-p}{2+p} \prod_{j=2}^{k} \frac{1+pj}{2+pj}, \quad k \geq 1.$$

In terms of the Gamma function,

$$P_k = \frac{(1-p)\,\Gamma(2+2/p)}{(2+p)\,\Gamma(2+1/p)} \cdot \frac{\Gamma(k+1+1/p)}{\Gamma(k+1+2/p)}. \tag{5}$$

As $k \to \infty$, we have $\Gamma(k+1+1/p)/\Gamma(k+1+2/p) \sim k^{-1/p}$, and so $P_k \sim \lambda_p k^{-1/p}$, where $\lambda_p = (1-p)\,\Gamma(2+2/p)/[(2+p)\,\Gamma(2+1/p)]$. The degree distribution thus follows a power law with exponent $1/p$, matching the degree distribution of the BA model at $p = 1/3$.

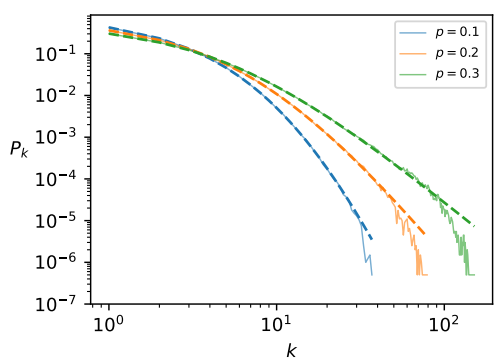

FIG. 2: Log-log scaled degree distribution P_k for $p = 0.1, 0.2, 0.3$, with $N = 10^3$ and averaging over 10^3 trials. Dashed lines show the analytical prediction (5).

V. TRIANGLES AND CLUSTERING

Let T denote the expected number of triangles in the network, and write T_N to specify when the network has N nodes. Triangles are created in an all-or-none network if and only if the "all" event occurs and $N \geq 2$. If the target u has degree k_u and there are e_u edges among its neighbors, the new node v forms k_u triangles of type (v, u, w_i) and e_u triangles of type (v, w_i, w_j), where w_i, w_j are distinct neighbors of u. Note that $\sum_u k_u = 2M_N$ and $\sum_u e_u = 3T_N$. Averaging over the target choice u thus yields

$$T_{N+1} = \left(1 + \frac{3p}{N}\right) T_N + \frac{2p\, M_N}{N}. \tag{6}$$

This linear recurrence has the same structure as (1). Substituting the exact edge formula and solving with $T_1 = 0$ gives, for $p \neq 1/2, 1/3$,

$$T = \frac{2pN}{(1-2p)(1-3p)} + \frac{2\mathcal{A}_N}{1-2p} - \frac{2\mathcal{B}_N}{1-3p}, \tag{7}$$

where $\mathcal{A}_N$ is as in (2) and $\mathcal{B}_N = \Gamma(N+3p)/[\Gamma(1+3p)\Gamma(N)]$. Thus two regimes emerge as $N \to \infty$: for $p < 1/3$, the growth $T \sim \frac{2p}{(1-2p)(1-3p)} N$ is linear, and for $p > 1/3$, the growth $T \sim \frac{2}{\Gamma(1+3p)(3p-1)} N^{3p}$ is superlinear.

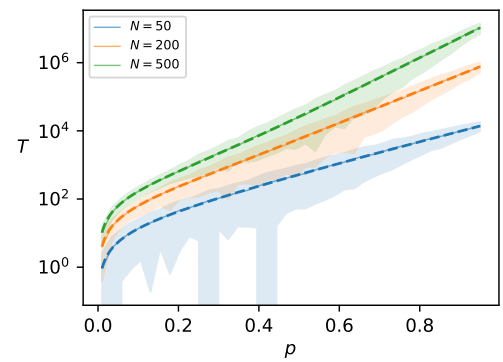

FIG. 3: Expected number of triangles T as a function of p for $N = 50, 200, 500$, averaged over 10^3 trials. Dashed lines show the analytical prediction (7).

We now derive the global clustering coefficient $C = 3T/W$, where $W = \frac{1}{2}\sum_i k_i(k_i - 1)$ is the expected number of connected triples. Let $S_N = \sum_i k_i^2$, and note that $W = (S_N - 2M_N)/2$. To derive a recurrence for S_N, note that when a node gains one link ($k \to k+1$), its contribution to S changes by $(k+1)^2 - k^2 = 2k+1$. At step $N \to N+1$, the target u always gains degree 1, and in the "all" event each neighbor w of u also gains degree 1. On the other hand, the new node enters with degree $k_v = 1$ or $1 + k_u$. Summing up all contributions, the expected change in S given target u is

$$\mathbb{E}[\Delta S \,|\, u] = (1-p) + p(1+k_u)^2 + (2k_u+1) + p\sum_{w\sim u}(2k_w+1),$$

Notice that

$$\sum_u \sum_{w\sim u} k_w = \sum_i k_i^2 = S, \tag{8}$$

as each node w appears as a neighbor of exactly k_w other nodes. Thus, averaging over the uniform choice of u, and using $\sum_u k_u = 2M$, $\sum_u k_u^2 = S$, and (8) gives

$$S_{N+1} = \left(1 + \frac{3p}{N}\right) S_N + 2 + \frac{(4+6p)M_N}{N}. \tag{9}$$

Solving (9) by the same method used for (1) and (6) gives, for $p \neq 0, \frac{1}{2}, \frac{1}{3}$,

$$S = \frac{(6+2p)N}{(1-2p)(1-3p)} + \frac{(4+6p)\mathcal{A}_N}{p(1-2p)} - \frac{4(1+2p)\mathcal{B}_N}{p(1-3p)}, \tag{10}$$

where $\mathcal{A}_N$ and $\mathcal{B}_N$ are as in (7). Recall that $W = (S - 2M)/2$, so substituting (2) and (10) yields

$$W = \frac{2(1+2p)}{p}\left[\frac{pN}{(1-2p)(1-3p)} + \frac{\mathcal{A}_N}{1-2p} - \frac{\mathcal{B}_N}{1-3p}\right],$$

Comparing with $3T$ from (7),

$$3T = 6\left[\frac{pN}{(1-2p)(1-3p)} + \frac{\mathcal{A}_N}{1-2p} - \frac{\mathcal{B}_N}{1-3p}\right],$$

we see that W and $3T$ have the same bracketed term, and thus we get

$$C = \frac{3p}{1+2p}, \tag{11}$$

for all $p \in (0,1)$ as $N \to \infty$. In particular, this result is constant on N, which is large compared to the clustering coefficient of the Erdős-Rényi random graph and thus a sign for the small-world property.

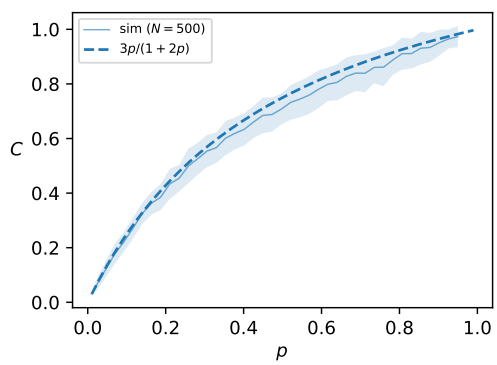

FIG. 4: Global clustering coefficient C as a function of p for $N = 500$ with 10^2 trials. Shaded regions show one standard deviation. Dashed line shows the analytical prediction (11).

VI. ASSORTATIVITY

Let X, Y be the random variables representing the degrees of two nodes of an edge chosen uniformly at random from the network. The degree assortativity coefficient [1] is the Pearson correlation coefficient of the degree of pairs of adjacent nodes:

$$r = \frac{\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]}{\sigma_X \sigma_Y} = \frac{\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}{\mathbb{E}[X^2] - (\mathbb{E}[X])^2}.$$

Note that $\mathbb{E}[X] = \mathbb{E}[Y] = \frac{1}{|E|}\sum_{(u,v)\in E}\frac{1}{2}(k_u + k_v)$, $\mathbb{E}[X^2] = \mathbb{E}[Y^2] = \frac{1}{|E|}\sum_{(u,v)\in E}\frac{1}{2}(k_u^2 + k_v^2)$, and $\mathbb{E}[XY] = \frac{1}{|E|}\sum_{(u,v)\in E} k_u k_v$. Thus we have

$$\begin{aligned} r &= \frac{\frac{1}{|E|}\sum_{(i,j)\in E} k_i k_j - \frac{1}{|E|^2}\left[\sum_{(i,j)\in E}\frac{1}{2}(k_i + k_j)\right]^2}{\frac{1}{|E|}\sum_{(i,j)\in E}\frac{1}{2}(k_i^2 + k_j^2) - \frac{1}{|E|^2}\left[\sum_{(i,j)\in E}\frac{1}{2}(k_i + k_j)\right]^2} \\ &= \frac{4MR - S^2}{2MQ - S^2}, \end{aligned} \tag{12}$$

where $M = |E|$ and $S = \sum_i k_i^2$ are defined as before, $Q = \sum_i k_i^3$, and $R = \sum_{(i,j)\in E} k_i k_j$. Let Q_N and R_N denote the corresponding quantities when the network has N nodes.

We first derive a recurrence for Q_N. Note that when a node i gains one edge ($k_i \to k_i + 1$), its contribution to Q changes by $(k_i+1)^3 - k_i^3 = 3k_i^2 + 3k_i + 1$. At step $N \to N+1$, the target u always gains degree 1, and in the "all" event each neighbor w of u also gains degree 1. On the other hand, the new node enters with degree $k_v = 1$ or $1 + k_u$. Summing up all contributions, the expected change in Q given target u is

$$\begin{aligned} \mathbb{E}[\Delta Q \mid u] = (1-p)\cdot 1 + p(1+k_u)^3 + (3k_u^2 + 3k_u + 1) \\ + p\sum_{w\sim u}(3k_w^2 + 3k_w + 1). \end{aligned}$$

By the same argument as (8), $\sum_u \sum_{w\sim u} k_w^2 = \sum_w k_w^3 = Q$. Thus, averaging over the uniform choice of u, and using $\sum_u k_u = 2M$, $\sum_u k_u^2 = S$, $\sum_u k_u^3 = Q$, and (8) yields

$$Q_{N+1} = \left(1 + \frac{4p}{N}\right)Q_N + 2 + \frac{(3+6p)S_N + (6+8p)M_N}{N}.$$

To derive a recurrence for R_N, note that when node i gains one edge, the product $k_i k_j$ on each incident edge (i,j) increases by k_j. If both endpoints gain degree simultaneously, the product increases by $k_i + k_j + 1$ instead. Write $\pi_i = \sum_{j\sim i} k_j$. We consider the two events separately.

"None" event. Only the target u gains degree 1, and the single new edge (v,u) contributes $k_v \cdot k_u^{\text{new}} = k_u + 1$. The total change is thus $\Delta R_{\text{none}} = \pi_u + k_u + 1$.

"All" event. The target u and all its neighbors gain degree 1, so the set of nodes whose degree increases is $\{u\} \cup N(u)$. Each node i that gains degree 1 increases the product $k_i k_j$ by k_j on every incident edge (i,j), contributing π_i in total, summing over all such nodes yields

$$\pi_u + \sum_{w\sim u}\pi_w. \tag{13}$$

Also note that, for an edge (i,j) where both endpoints gain degree 1, the change in the product is $(k_i+1)(k_j+1) - k_i k_j = k_i + k_j + 1$, but (13) counts only $k_i + k_j$. Thus each such edge requires an additional $+1$ correction. The edges with both endpoints in $\{u\} \cup N(u)$ are the k_u edges (u,w) and the e_u edges among neighbors of u, so we need to add $k_u + e_u$ to the correction. For the new

edges, the new node has degree $k_v = 1+k_u$, so their total contribution to R is

$$(k_u+1)^2 + \sum_{w\sim u}(k_u+1)(k_w+1) = (k_u+1)(2k_u+1+\pi_u).$$

Combining all three parts, we get

$$\Delta R_{\text{all}} = (2+k_u)\pi_u + \sum_{w\sim u}\pi_w + 2k_u^2 + 4k_u + 1 + e_u.$$

Thus the expected change in R given target u is $\mathbb{E}[\Delta R \mid u] = (1-p)\,\Delta R_{\text{none}} + p\,\Delta R_{\text{all}}$. Note that

$$\sum_u \sum_{w\sim u} \pi_w = \sum_w k_w\,\pi_w = 2R, \tag{14}$$

as each edge $\{w,z\}$ contributes $k_w k_z$ once on each endpoint. Averaging over the uniform choice of u using (14), $\sum_u \pi_u = S$, $\sum_u k_u\pi_u = 2R$, $\sum_u e_u = 3T$, $\sum_u k_u = 2M$, and $\sum_u k_u^2 = S$ yields

$$R_{N+1} = \left(1+\frac{4p}{N}\right) R_N + 1 + \frac{(1+3p)S_N + (2+6p)M_N + 3pT_N}{N}.$$

Recurrences for both Q and R share the homogeneous growth factor $(1+4p/N)$, with solution $\mathcal{C}_N = \Gamma(N+4p)/[\Gamma(1+4p)\,\Gamma(N)]$. Substituting the exact formulas for M_N, S_N, T_N from Sections III–V into the forcing terms and solving by the same method used for (1),(6), and (9) yields, for $p \neq 0, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}$,

$$\begin{aligned} Q_N &= \frac{(26+22p)N}{(1-2p)(1-3p)(1-4p)} - \frac{(12+36p+28p^2)\mathcal{A}_N}{2p^2(1-2p)} \\ &\quad + \frac{4(3+12p+12p^2)\mathcal{B}_N}{p^2(1-3p)} + \lambda_Q\,\mathcal{C}_N, \\ R_N &= \frac{3(3+5p)N}{(1-2p)(1-3p)(1-4p)} - \frac{(4+16p+18p^2)\mathcal{A}_N}{2p^2(1-2p)} \\ &\quad + \frac{2(2+10p+15p^2)\mathcal{B}_N}{p^2(1-3p)} + \lambda_R\,\mathcal{C}_N, \end{aligned}$$

where $\mathcal{A}_N$ and $\mathcal{B}_N$ are as in (7), and λ_Q, λ_R are constants determined by the initial conditions $Q_1 = R_1 = 0$.

Since $\mathcal{C}_N \sim N^{4p}$, for $p < 1/4$, the homogeneous terms $\mathcal{A}_N$, $\mathcal{B}_N$, $\mathcal{C}_N$ are all sublinear and all five quantities M, S, T, Q, R grow linearly. Let

$$\alpha_Q = \frac{26+22p}{(1-2p)(1-3p)(1-4p)},\ \alpha_R = \frac{3(3+5p)}{(1-2p)(1-3p)(1-4p)}.$$

be the linear term coefficients of Q_N and R_N respectively. From (1) and (10) we can also see that $\alpha_M = 1/(1-2p)$ and $\alpha_S = (6+2p)/[(1-2p)(1-3p)]$. Substituting into (12) and simplifying, we get that, for $p < 1/4$, as $N \to \infty$,

$$r \sim \frac{4\alpha_M\alpha_R}{2\alpha_M\alpha_Q - \alpha_S^2} = \frac{p(9-2p)}{(1+2p)(2-p)}. \tag{15}$$

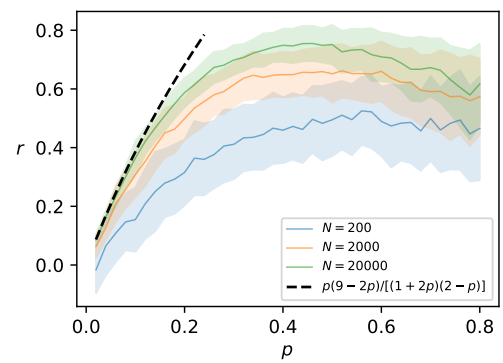

FIG. 5: Assortativity coefficient r as a function of p for $N = 200, 2000, 20000$ with 10^2 trials. Shaded regions show one standard deviation. The dashed line show the asymptotic prediction (15) for $p < 1/4$. The simulations slowly converge to the predicted curves, with higher values of p exhibiting even slower convergence.

VII. AVERAGE SHORTEST-PATH DISTANCE

Let $D = \sum_{i<j} d(i,j)$ denote the expected sum of all pairwise shortest-path distances, and write D_N for this sum when the network has N nodes. The average shortest-path distance is then

$$\ell = \frac{2D_N}{N(N-1)}. \tag{16}$$

At step $N \to N+1$, a target u is chosen uniformly from the N existing nodes. Note that adding v does not shorten any existing pairwise distance, as any path through v can be rerouted through u or its neighbors without increasing length. Thus $D_{N+1} = D_N + \sum_j d(v,j)$. Write $\sigma_u = \sum_{j\neq u} d(u,j)$. We now consider the two events separately.

"None" event. The new node v is connected only to the target u. Thus, $d(v,u) = 1$ and $d(v,j) = 1 + d(u,j)$ for every $j \neq u$, which yields $\sum_j d(v,j) = N + \sigma_u$.

"All" event. The new node v is adjacent to the target u and all its neighbors, so any path from u to j can be rerouted through v by simply replacing the starting point u with v. Any path from v to j may also be modified to a path from u to j in the same fashion. Thus, $d(v,j) = d(u,j)$ for all $j \neq u$, and so $\sum_j d(v,j) = 1 + \sigma_u$.

Averaging over the two events and the uniform choice of u, and noting that $\frac{1}{N}\sum_u \sigma_u = 2D_N/N$, we have $\sum_j d(v,j) = (1-p)N + p + 2D_N/N$, and so

$$D_{N+1} = \left(1+\frac{2}{N}\right) D_N + (1-p)N + p. \tag{17}$$

The homogeneous solution is $\mathcal{D}_N = N(N+1)/2$, satisfying $\mathcal{D}_{N+1}/\mathcal{D}_N = (N+2)/N$. Set $D_N = \gamma_N \mathcal{D}_N$. Then (17) is reduced to

$$\gamma_{N+1} - \gamma_N = \frac{2[(1-p)N+p]}{(N+1)(N+2)} = \frac{2(2p-1)}{N+1} + \frac{2(2-3p)}{N+2}.$$

It follows from $\gamma_1 = 0$ that

$$\gamma_N = \sum_{k=1}^{N-1} \frac{2(2p-1)}{k+1} + \frac{2(2-3p)}{k+2}$$
$$= 2\big((2p-1)(H_N-1) + (2-3p)(H_{N+1}-\tfrac{3}{2})\big)\,, \quad (18)$$

where $H_N = \sum_{j=1}^{N} 1/j$ is the N-th harmonic number. Since $H_N \sim \ln N$ as $N \to \infty$, we have $\gamma_N \sim 2(1-p)\ln N$, and so $D_N \sim (1-p)\,N(N+1)\ln N$. Substituting into (16) now yields

$$\ell \sim 2(1-p)\ln N,$$

for all $p \in [0,1)$. At $p = 1$ the network is a complete graph with $\ell = 1$. The distance scales logarithmically for $p \in (0,1)$ with no phase transition, confirming the small-world property we established with the constant clustering coefficient from Section V.

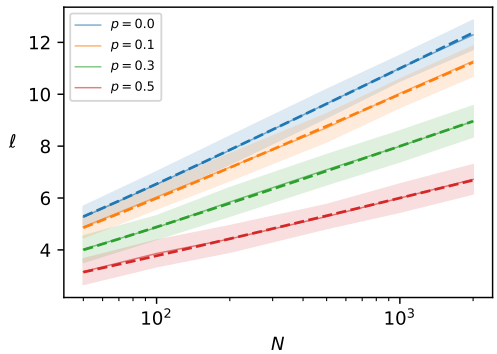

FIG. 6: Average shortest-path distance ℓ as a function of log scaled N for $p = 0.0, 0.1, 0.3, 0.5$. Dashed lines show the analytical prediction (18), fitting the simulation results perfectly across all tested p. Note that we use (18) in the figure instead of the asymptotic formula $2(1-p)\ln N$ to retain the constant correction terms.

VIII. CONSENSUS DYNAMICS

The consensus dynamics of a network is the process of all nodes reaching a common state. Assign each node with a scalar state $x_i \in \mathbb{R}$ and let $L = D - A$ be the Laplacian matrix of the network. The consensus dynamics is defined as $\dot{\mathbf{x}} = -L\mathbf{x}$. Let $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_N$ be the eigenvalues of L, and let v_i be the corresponding eigenvectors. Then the consensus dynamics converges to a mean state $\frac{1}{N}\sum_i x_i(0)\mathbf{1}$ at a rate determined by the second smallest eigenvalue λ_2. Specifically, $\mathbf{x}(t) = \sum_i e^{-\lambda_i t}\mathbf{v}_i\mathbf{v}_i^T\mathbf{x}(0)$, and so the contribution of $e^{-\lambda_i t}\mathbf{v}_i\mathbf{v}_i^T\mathbf{x}(0)$ decays exponentially with rate λ_i for each i. Since λ_2 is the smallest non-zero eigenvalue, the second term decays the slowest and thus it sets the convergence time $\tau = 1/\lambda_2$. In [2], it is shown that $\lambda_2 \ge 1/(N \cdot \mathrm{diam}(G))$, where $\mathrm{diam}(G)$ is the diameter of the network. Note that the all-or-none network is a random recursive tree at $p = 0$, which has diameter $O(\log N)$ [3]. Since the diameter would only decrease as p increases, we have $\lambda_2 = \Omega\,(1/N\log N)$. It follows that $\tau = O(N\log N)$. However, this bound is likely not tight, and the *Cheeger inequality* could yield a stronger estimate of λ_2 if the *conductance* could be bounded directly. We do not pursue this here. On the other hand, the network is a complete graph at $p = 1$, which has Laplacian matrix $L = (N-1)I - A$. In particular, $A = \mathbf{1}\mathbf{1}^T - I$, so $L = N \cdot I - \mathbf{1}\mathbf{1}^T$, which has eigenvalue 0 with multiplicity 1 and eigenvalue N with multiplicity $N-1$. Thus $\lambda_2 = N$ and the timescale for consensus is $\tau = 1/N$ at $p = 1$. This implies that τ collapses from $O(N\log N)$ to $1/N$ as p increases from 0 to 1, as shown in Figure 7.

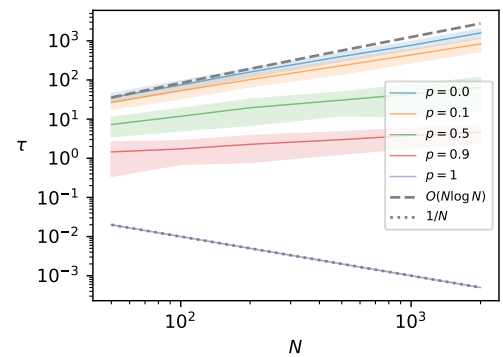

FIG. 7: As $p \to 0$, τ approaches the $O(N\log N)$ scaling of the random recursive tree. For $p > 0$, τ grows considerably slower than $N\log N$, suggesting the diameter-based bound is not tight. At $p = 1$ the network is complete and $\tau = 1/N$.

IX. CONCLUSION

We have presented an analytical study of the all-or-none copying model, showing that its single parameter p governs a rich variety of network phenomena. The edge count exhibits a sparse-to-dense transition at $p = 1/2$, while the degree distribution follows a power law $P_k \sim k^{-1/p}$, recovering the exponent in the BA model at $p = 1/3$ without invoking preferential attachment. A clustering coefficient $C = 3p/(1+2p)$ independent to N and logarithmic distances $\ell \sim 2(1-p)\ln N$ establish the small-world property for every $p \in (0,1)$, and a positive assortativity $r = p(9-2p)/[(1+2p)(2-p)]$ for $p < 1/4$. For the network dynamics, our logarithmic diameter bound yields a consensus timescale of $\tau = O(N\log N)$ for general p, while the complete-graph limit at $p = 1$ gives $\tau = 1/N$, confirming that a minimal copying rule can produce realistic network structure and efficient dynamics simultaneously.

[1] M. E. J. Newman, Phys. Rev. Lett. **89**, 208701 (2002).
[2] K. Das, Computers & Mathematics with Applications **48**, 715 (2004).
[3] B. Pittel, Random Structures & Algorithms **5**, 337 (1994), https://onlinelibrary.wiley.com/doi/pdf/10.1002/rsa.3240050207.